① EXPRESS THE AREA "A" OF AN EQUILATERAL TRIANGLE AS A FUNCTION of its SIDE "S".

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② GIVEN $g(t) = \sqrt{t + 1.0604} - 6t^3$

FIND $g(0.9261)$

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③ FIND THE DOMAIN AND RANGE of THE FUNCTION GIVEN BELOW:

$$T(t) = 2t^4 + t^2 - 1$$

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④ FIND THE DOMAIN of THE FUNCTION

$$g(x) = \frac{\sqrt{x-2}}{x-3}$$

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⑤ EXPRESS THE Cost "C" of INSULATING A CYLINDRICAL WATER TANK of HEIGHT 2 m AS A FUNCTION of its RADIUS "r", if THE Cost of INSULATION is \$3 PER SQUARE METER.

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⑥ A CITY DOES NOT TAX THE FIRST \$30,000.00 of A RESIDENT's INCOME BUT TAXES ANY AMOUNT OVER \$30,000 at 5%.

a) FIND THE TAX T AS A FUNCTION of THE RESIDENT's INCOME I.

b) FIND $f(25000)$ & $f(45000)$.

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(7) THE TOTAL ANNUAL FRACTION "f" OF ENERGY SUPPLIED BY SOLAR ENERGY TO A HOME AS A FUNCTION OF THE AREA "A" (in m^2) OF SOLAR COLLECTOR IS

$f = 0.065\sqrt{A}$, PLOT f AS A FUNCTION OF "A"

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(8) PLOT THE GRAPH

$$f(x) = \begin{cases} \dfrac{1}{x-1} & (\text{for } x < 0) \\ \sqrt{x+1} & (\text{for } x \geq 0) \end{cases}$$

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(9) PLOT $y = \dfrac{x^2}{\sqrt{3-x}}$

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(10) THE CUTTING SPEED "S" (in ft/min) OF A SAW IN CUTTING A PARTICULAR TYPE OF METAL PIECE IS GIVEN BY $S = \sqrt{t - 4t^2}$, WHERE t IS THE TIME IN SECONDS. WHAT IS THE MAXIMUM CUTTING SPEED IN THIS OPERATION?

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(11) THE TIME REQUIRED FOR A SUM OF MONEY TO DOUBLE IN VALUES, WHEN COMPOUNDED ANNUALLY, IS GIVEN AS A FUNCTION OF THE INTEREST RATE IN THE FOLLOWING TABLE:

RATE %	4	5	6	7	8	9	10
TIME (YEARS)	17.7	14.2	11.9	10.2	9	8	7.3

FIND THE RATE FOR $t = 10.0$ YEARS.

(12) IF THE FUNCTION $y = 3 - 2x$ IS SHIFT RIGHT 1 AND DOWN 3, WHAT IS THE RESULTING FUNCTION?

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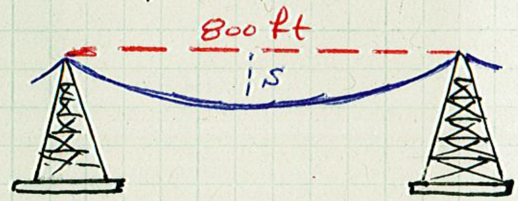
(13) THE VERTICAL SAG "S" (IN FT) AT THE MIDDLE OF AN 800-FT POWER LINE AS A FUNCTION OF TEMPERATURE T (IN °F) IS GIVEN IN THE FOLLOWING TABLE: (SHOWN) FOR THE FUNCTION $S = f(T)$, FIND $f(47)$ BY LINEAR INTERPOLATION

T (°F)	0	20	40	60
S (FT)	10.2	10.6	11.1	11.7



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(14) A FOUNTAIN IS AT THE CENTER OF A CIRCULAR POOL OF RADIUS "r" FT, AND THE AREA IS ENCLOSED WITHIN A CIRCULAR RAILING THAT IS 45 FT FROM THE EDGE OF THE POOL. IF THE TOTAL AREA WITHIN THE RAILING IS 11,500 ft^2, WHAT IS THE RADIUS OF THE POOL?

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(15) AN ACTUATOR USED IN A PROSTHETIC ARM CAN PRODUCE A DIFFERENT AMOUNT OF FORCE BY CHANGING THE VOLTAGE OF THE POWER SUPPLY. THE FORCE AND VOLTAGE SATISFY THE LINEAR RELATION $F = KV$, WHERE K IS THE VOLTAGE APPLIED AND F IS THE FORCE PRODUCED BY THE PROSTHETIC ARM. THE MAXIMUM FORCE THE ARM CAN PRODUCE IS $F = 44.5$ N WHEN SUPPLIED WITH $V = 12$ VOLTS.

FIND THE FORCE PRODUCED BY THE ACTUATOR WHEN SUPPLIED WITH $V = 7.3$ VOLTS.